Milk Filter Machine



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Milk Filter machine

What it is used for

To remove visible suspended particles from raw milk using a milk filter machine before further dairy processing. To maintain hygienic milk handling and ensure that the filter is correctly installed and replaced when needed.

Main parts

- Machine body and milk inlet
- Filter housing and filter element
- Milk outlet and collection vessel
- Pump or gravity-feed arrangement, if fitted
- Valves and flow controls
- Cover, seals, and support frame

Before you start

1. Confirm that the filter machine is designed for the intended milk flow and operating conditions.
2. Clean and sanitize all milk-contact surfaces using the approved dairy hygiene procedure.
3. Inspect the filter housing, seals, valves, pipes, and collection vessel for damage or residue.
4. Install the correct food-grade filter element and check that it is seated properly.
5. Ensure that milk containers are clean, covered where appropriate, and ready for use.
6. Check that valves are positioned correctly and that the outlet path is clear.

How to use

1. Keep the milk supply isolated while opening the filter housing or installing the filter element.
2. Fit the filter element in the correct direction and close the housing securely.
3. Position a clean collection vessel under the outlet or connect the approved downstream line.
4. Transfer raw milk into the inlet vessel using hygienic handling practices.
5. Open the inlet valve gradually or start the pump according to the machine instructions.
6. Monitor milk flow and inspect the filter housing and connections for leakage.
7. Check the outlet milk and filter condition during operation without opening a pressurized system.
8. Stop the flow if it becomes unusually slow, the filter blocks, or leakage occurs.
9. Isolate the supply and release pressure according to the approved procedure before opening the housing.
10. Remove the used filter element carefully and discard or clean it according to the filter type and food-safety procedure.

1. Clean and sanitize the housing, seals, pipes, and collection vessel after processing.
2. Record filter replacement, cleaning, and any abnormal observations.

⚠ Safety

- Use only food-grade filter elements and materials suitable for milk contact.
- Maintain cold-chain and milk-handling requirements specified by the dairy process.
- Never open a pressurized filter housing or disconnect a line while it is under pressure.
- Keep hands and tools clean and prevent contamination of filtered milk.
- Do not continue operation with damaged seals, cracked housings, or leaking connections.
- Follow the approved procedure for handling and disposing of used filters and contaminated residues.

Care & storage

- Clean and sanitize all milk-contact components after each processing session or batch as required.
- Inspect filter elements, seals, clamps, and valves regularly.
- Replace disposable filter elements at the specified interval or when blocked or damaged.
- Check pumps and flow controls where fitted and maintain them according to the manual.
- Store clean, dry parts in a protected hygienic area.

Common mistakes to avoid

- Installing the filter element in the wrong direction or leaving it poorly seated.
- Using a non-food-grade or damaged filter element.
- Opening the housing while it is pressurized.
- Allowing dirty tools or containers to contaminate filtered milk.
- Continuing to use a clogged filter and ignoring reduced flow or leakage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.